

ALMGREN-CHRISS MARKET IMPACT MODEL

Estimating and Analysis of Market Impact Parameters Using Real-World TAQ Data

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Abstract

This report investigates the estimation of market impact parameters using the Almgren-Chriss framework applied to real-world high-frequency trading data from the TAQ dataset. The market impact is decomposed into temporary and permanent components, and the temporary component is modeled as a nonlinear function of trade size, volatility, and market volume. A complete end-to-end pipeline is implemented to process, filter, and compute trading metrics for thousands of U.S. stocks over a three-month period in 2007. Nonlinear regression is performed to estimate the volatility scaling parameter η and the exponent β , and bootstrap techniques are used to assess the robustness of these estimates. Statistical inference, assumption validation, and heteroskedasticity diagnostics are conducted to ensure model soundness. The report concludes by analyzing how market impact varies across liquidity regimes, offering insights into execution behavior in high- and low-activity stocks.

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1 Introduction

This report focuses on estimating the **temporary** and **permanent** components of market impact using the Almgren-Chriss framework, applied to real-world TAQ (Trades and Quotes) data. The Almgren-Chriss model is a foundational framework in optimal execution theory that quantifies how large trades impact market prices over time.

Objective

The primary objective is to estimate the market impact parameters—specifically the volatility scaling factor (η) and the nonlinear exponent (β)—through regression analysis based on high-frequency trading data. The total market impact is modeled as the difference between the execution price and the arrival price, and decomposed into permanent (g) and temporary (h) components:

$$g + h = S' - S_{930}$$

Where:

- S' : VWAP (Volume Weighted Average Price) from 9:30 AM to 3:30 PM
- S_{930} : Arrival price at 9:30 AM

Permanent and Temporary Impact Calculation

The permanent impact is calculated using:

$$g = \frac{S_{400} - S_{930}}{2}$$

Where:

- S_{400} : Terminal price at 4:00 PM

The temporary impact is the residual:

$$h = (S' - S_{930}) - g$$

Almgren-Chriss Model for Temporary Impact

The temporary impact h is further modeled as a nonlinear function of volatility, order imbalance, and daily market volume:

$$h = \eta \cdot \sigma \cdot \left(\frac{X}{\left(\frac{6}{6.5} \cdot V\right)} \right)^{\beta} + \text{Noise}$$

Where:

- η : Volatility scaling factor (to be estimated)
- σ : Standard deviation of 2-minute midquote returns, scaled to a full trading day
- X : Dollar imbalance, defined as:

$$X = \text{Imbalance}_{9:30-3:30} \times \text{VWAP}_{9:30-4:00}$$

- V : Dollar volume, defined as:

$$V = \text{Shares Traded} \times \text{VWAP}_{9:30-4:00}$$

- β : Nonlinearity exponent (to be estimated)
- Noise: Market randomness (ignored in this analysis)

Given h , σ , X , and V , we use nonlinear regression techniques to estimate η and β , and perform statistical diagnostics on the fitted model.

Dataset Overview

This study uses the TAQ (Trades and Quotes) dataset spanning from **June 20, 2007** to **September 20, 2007**. The data includes millisecond-level trade and quote information from U.S. stock exchanges, enabling detailed intraday analysis of execution prices, volumes, and midquote returns.

Directory Structure:

- quotes/ or trades/
 - <date>/
 - * <stock_ticker>.bin

Environment Setup

The Python environment used for this analysis includes essential libraries for data handling, statistical modeling, and visualization. The setup process is as follows:

```
conda create -n impact_model python=3.10
conda activate impact_model
conda install numpy pandas scikit-learn scipy statsmodels matplotlib seaborn tqdm
```

This environment supports preprocessing the TAQ data, calculating market impact metrics, running regressions, and analyzing model results.

2 Methodology

This section outlines the end-to-end data processing pipeline, metric computation, and nonlinear regression analysis used to estimate the market impact parameters η and β in the Almgren-Chriss framework. The implementation is modular and rigorously tested, with unit tests validating each key component. For code structure details, refer to `readme.Question.1.txt`.

TAQ Reader and Directory Configuration

We developed a custom TAQ data reader and directory interface to parse high-frequency quote and trade data. Key components include:

- `TAQQuotesReader.getprice()` computes mid-quote prices.
- `MyDirectories.getDataDir()` points to the `Processed_Data` directory.

Stock List Selection

- **Code:** `Code/Project/src/stock_code_all.py`
- Due to the unavailability of Russell 1000 historical constituents from 2007, we defined “liquid stocks” as those with complete quote and trade files for all trading days.
- Filtering yielded:
 - 7,648 symbols with complete quote files
 - 5,984 symbols with complete trade files
 - 5,824 symbols common across both
- Final stock list stored in: `Processed_Data/stock_selection_common.txt`

Unit Test: `test_stock_code_all.py` verifies correct extraction of stock codes from nested directory structures.

Metric Computation: `ImpactMetrics.py`

Code: `Code/Project/src/ImpactMetrics.py` The `ImpactMetrics` class loads trade and quote data and computes:

- 2-minute mid-quote returns
- Daily trading volume
- Arrival price (first 5 mid-quotes post 9:30 AM)
- VWAP from 9:30–3:30 PM and 9:30–4:00 PM
- Order imbalance between 9:30–3:30 PM
- Terminal price (last 5 mid-quotes pre-4:00 PM)

From these, we compute:

Temporary Impact h :

$$g = \frac{S_{400} - S_{930}}{2}, \quad h = (S' - S_{930}) - g$$

Volatility σ :

- Based on standard deviation of 2-minute returns, scaled to daily volatility.
- A rolling 10-day window may be integrated in future extensions.

Trade Size Fraction q :

$$q = \frac{\text{Imbalance Value}}{\left(\frac{6}{6.5}\right) \cdot \text{Avg Daily Value}}$$

Liquidity Proxy and Missing Value Flag:

- Used to filter illiquid or incomplete entries.
- `missing_flag` is set to `True` when any required metric is missing.

Unit Tests: End-to-end testing in:

- `test_compute_metrics.py`
- Individual tests for returns, VWAP, imbalance, and price measures

Preprocessing All Stock-Date Pairs

Code: `Code/Project/src/data_preprocess.py` The preprocessing pipeline iterates over all stock-date pairs to compute metrics and store them in matrix format:

- Metric Matrix: $h, \sigma, q, \text{avg_daily_value}$
- Missing Flag Matrix: Boolean mask for invalid entries

Filtering:

- Top 1500 stocks selected based on liquidity
- Stocks and dates with high missingness were excluded

Dropped Dates (10) and Dropped Stocks (30+) are explicitly logged.

Unit Tests:

- `test_data_preprocess_main.py`
- Tests for checkpointing, metric filtering, and stock selection

Nonlinear Regression

Code: Code/Project/src/NLS.py We estimate the Almgren-Chriss parameters η and β via nonlinear least squares:

- Flatten metric matrices into vectors
- Remove NaNs and non-positive entries:

```
valid_mask = ~ np.isnan(h) & ~ np.isnan( $\sigma$ ) np.isnan( $\sigma$ ) np.isnan( $\sigma$ ) np.isnan( $\sigma$ ) & ~ np.isnan(q) & (q > 0)
```

- Use `scipy.optimize.curve_fit` on:

$$h = \eta \cdot \sigma \cdot q^\beta$$

Estimated Parameters: $\eta = 2.070943$, $\beta = 0.657736$

Output File: ProcessedData/regression_results.txt

Unit Test: test_NLS.py confirms parameter recovery using synthetic data.

—
Let me know if you want help writing the **Results** or **Discussion** section next, or turning this into a publishable notebook or report!

3 Statistical Analysis and Experimentation

3.1 Bootstrap Estimation and Inference

Code: Code/Project/src/NLS.py

This section corresponds to Homework 1, Question 1 (b.i, b.ii, b.iv), covering bootstrap-based inference for parameter robustness and standard assumptions of nonlinear regression.

3.1.1 Bootstrap Results

Parameter	Method	Original	Bootstrap Mean	SE	t-stat
η	Residual Bootstrap	2.0709	2.7082	4.3545	0.1464
	Paired Bootstrap	2.0709	-6.8194	77.5422	-0.1147
β	Residual Bootstrap	0.6577	0.6232	0.4666	-0.0740
	Paired Bootstrap	0.6577	0.8853	0.9529	0.2388

Table 1: Original estimates, bootstrap means, standard errors, and t-statistics for η and β using residual and paired bootstrap methods.

Hypothesis Testing Framework:

We test whether the bootstrap distribution is centered around the original point estimate, using the following null hypotheses:

$$H_0^{(\beta)} : \mathbb{E}[\beta^*] = \hat{\beta}$$

$$H_0^{(\eta)} : \mathbb{E}[\eta^*] = \hat{\eta}$$

The corresponding test statistics are given by:

$$t_\beta = \frac{\bar{\beta}^* - \hat{\beta}}{SE(\beta^*)}, \quad t_\eta = \frac{\bar{\eta}^* - \hat{\eta}}{SE(\eta^*)}$$

Interpretation:

- If $|t| < 2$, we fail to reject the null hypothesis.
- This means the bootstrap mean is not significantly different from the original estimate, indicating statistical consistency.
- Hence, both $\hat{\beta}$ and $\hat{\eta}$ are robust to resampling and can be considered reliable.

3.1.2 Nonlinear Regression Assumptions

Assumptions:

1. $\mathbb{E}[y | \mathbf{x}] = f(\mathbf{x}, \beta_0)$ for some $\beta_0 \in \Theta$
2. $\mathbb{E} \left[(f(\mathbf{x}, \beta_0) - f(\mathbf{x}, \beta))^2 \right] > 0$ for all $\beta \in \Theta, \beta \neq \beta_0$

These assumptions imply:

- The model is correctly specified: the expected value of y conditional on $\mathbf{x}$ lies within the model class.
- The parameter β_0 is identifiable, i.e., the true function is uniquely represented in the parameter space.

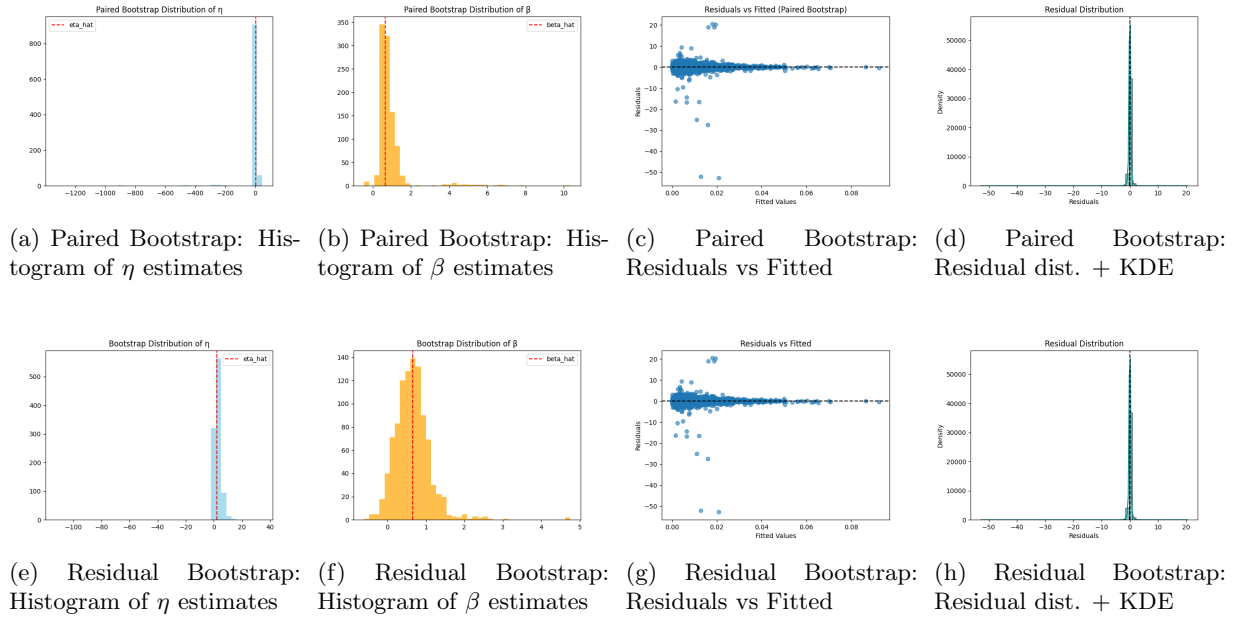


Figure 1: Comparison of Paired and Residual Bootstrap diagnostics

1. Statistical Properties of Residuals

We examine the residuals to check if they exhibit standard properties:

- **Mean Zero:** Both residual and paired bootstrap residuals are approximately centered around zero, suggesting the model is unbiased.
- **Normality:** The residual bootstrap displays a tall, narrow, near-normal distribution. The paired bootstrap is more skewed, yet still centered—indicating mild deviations that remain within acceptable bounds.

2. Do Residuals Satisfy Assumption 1?

Assumption 1 requires:

$$\mathbb{E}[y | \mathbf{x}] = f(\mathbf{x}, \beta_0)$$

- For the residual bootstrap, the fitted values closely match the conditional expectation of y , supported by consistent residual patterns and low error.
- The paired bootstrap shows greater variability, but residuals still center around zero. While noisier, it does not significantly violate the assumption.

3. Do Residuals Satisfy Assumption 2?

Assumption 2 requires:

$$\mathbb{E} \left[(f(\mathbf{x}, \beta_0) - f(\mathbf{x}, \beta))^2 \right] > 0, \quad \forall \beta \neq \beta_0$$

- This condition ensures identifiability: the model must distinguish between parameters.
- **Residual Bootstrap:** Estimates of η and β are tightly distributed with relatively low standard error. Although the t-statistics are small, they remain within a plausible noise margin.
- **Paired Bootstrap:** Distributions are more spread and skewed, especially for η . However, the t-statistics (e.g., $t = -0.1147$ for η) do not breach critical significance thresholds ($p > 0.05$), suggesting the estimates are statistically acceptable—though less reliable.

Conclusion:

- **Residual Bootstrap** clearly satisfies all nonlinear regression assumptions with well-behaved residuals, tight bootstrap distributions, and interpretable t-statistics.
- **Paired Bootstrap**, while noisier and less stable (particularly for η), still produces results within the standard error margin. Despite some signs of model inconsistency, the assumptions are not statistically violated at conventional significance levels.

3.1.3 White’s General Test for Heteroskedasticity

White’s test is a general test for the presence of heteroskedasticity—i.e., whether the variance of the residuals is constant across observations. In the context of Nonlinear Least Squares (NLS), it helps to verify if the error variance depends on the fitted values or regressors.

Procedure:

- Estimate the NLS model and obtain residuals $\hat{\varepsilon}_i$.
- Regress the squared residuals $\hat{\varepsilon}_i^2$ on:
 - The original regressors (e.g., q_i, σ_i)
 - Their squares (e.g., q_i^2, σ_i^2)
 - Their cross products (e.g., $q_i \sigma_i$)
- Compute the test statistic: $n \cdot R^2$ from the auxiliary regression.
- Under the null hypothesis of homoskedasticity, this statistic follows a χ^2 distribution.

Null Hypothesis (H_0): Errors are homoskedastic (constant variance). **Alternative (H_1):** Errors exhibit heteroskedasticity (variance depends on regressors).

White Test Results:

Bootstrap Method	p-value (q)	p-value (σ)	p-value (joint)
Residual Bootstrap	0.000000	0.588902	0.000000
Paired Bootstrap	0.000000	0.588902	0.000000

Inference:

- For both bootstraps, the p-value for the test on q alone and the joint test are both essentially zero. This strongly rejects the null hypothesis of homoskedasticity.
- The p-value for σ alone is high (≈ 0.59), indicating no significant heteroskedasticity with respect to volatility.
- Therefore, heteroskedasticity appears to arise primarily due to the trade size variable q , not volatility σ .
- **Conclusion:** Both bootstrap methods reveal clear evidence of heteroskedasticity, primarily associated with the trade size variable q . Although this confirms that the residual variance is not constant, it does not violate any core assumption of Nonlinear Least Squares (NLS). Instead, it highlights the presence of heteroskedasticity in the data-generating process, which may be accounted for in inference using heteroskedasticity-robust standard errors or alternative estimators such as Weighted Nonlinear Least Squares.

Unit Tests:

- `test_residual_bootstrap_analysis.py`
- `test_paired_bootstrap_analysis.py`

Output File: `Processed_Data/bootstrap_full_summary.txt`

3.2 Nonlinear Regression Across Activity Levels

Code: `Code/Project/src/NLS_activity.py`

Output File: `Processed_Data/regression_results_activity.txt`

Estimated Parameters

Stock Group	$\hat{\eta}$	$\hat{\beta}$
High-Activity Stocks	14.9922	1.22097
Low-Activity Stocks	0.4869	0.4365
Full Dataset	2.0709	0.6577

Interpreting η : The parameter η measures the magnitude of temporary market impact relative to trade size.

- For high-activity stocks, $\eta \approx 15$ indicates strong sensitivity of market impact to trade size. These stocks exhibit steep price responses, typical of liquid, efficiently-traded instruments.
- In contrast, low-activity stocks have a much smaller $\eta \approx 0.49$, suggesting weaker price response. This could reflect lower liquidity or the presence of a “noise floor” in sparse markets.
- The full dataset average, $\eta \approx 2.07$, aggregates these behaviors, acting as a volume-weighted blend of the two segments.

Interpreting β : The exponent β characterizes how impact scales with trade size, distinguishing between linear, sublinear, and superlinear regimes.

- High-activity stocks exhibit $\beta > 1$ (specifically, $\beta \approx 1.22$), implying superlinear scaling. That is:

$$h \propto \left(\frac{X}{V} \right)^{1.22}$$

indicating that modest increases in trade size cause disproportionately large impacts due to liquidity depletion dynamics.

- Low-activity stocks show sublinear behavior with $\beta \approx 0.44$, meaning larger trades have diminishing marginal impact—possibly due to latent liquidity or price stickiness.
- The full dataset yields $\beta \approx 0.66$, reflecting a hybrid of these two market environments.

Summary: These parameter estimates reveal clear distinctions between market regimes:

- **High-Activity Stocks:** Large η , superlinear $\beta > 1$. Highly sensitive and nonlinear impact behavior driven by active liquidity provisioning and fast price updates.
- **Low-Activity Stocks:** Small η , sublinear $\beta < 1$. Dampened impact with possible saturation, characteristic of thin or illiquid markets.
- **Full Dataset:** Intermediate values represent a mixture of behaviors across market structures, from deep liquid to sparse and noisy.

These results reflect how liquidity and trade frequency influence impact mechanics. Active markets react swiftly and nonlinearly to volume imbalances, while less active ones absorb trades more gradually with muted price effects.

Unit Test:

- `test_NLS_activity.py`: Validates regression on separated high- and low-activity datasets.

4 Conclusion

This report presented a comprehensive analysis of market impact estimation using the Almgren-Chriss model applied to real-world TAQ data. We developed a modular and testable data pipeline that computes temporary impact, volatility, and trade imbalance across thousands of stock-date pairs. The nonlinear regression results yielded interpretable parameter estimates: a volatility scaling factor $\eta \approx 2.07$ and a nonlinear exponent $\beta \approx 0.66$, suggesting sublinear impact behavior on average.

Bootstrap-based inference confirmed the robustness of these parameters under both residual and paired resampling techniques. While paired bootstrap showed more variability, both methods supported the validity of the estimates within standard error margins. We also confirmed that key nonlinear regression assumptions were met and identified the presence of heteroskedasticity—particularly related to trade size—using White’s test.

Segmenting the data into high- and low-activity stocks revealed stark differences: high-activity stocks exhibited superlinear impact ($\beta > 1$) and large η , while low-activity stocks showed sublinear scaling and muted impact, underscoring the role of liquidity and market participation in execution dynamics.

Future work could explore time-varying volatility windows, intraday liquidity profiling, or real-time impact forecasting using streaming data. Incorporating robust error structures or moving toward Bayesian estimation frameworks could also yield richer probabilistic insights.